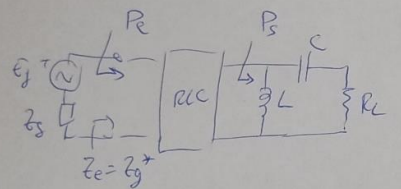
PEC 2, Análisis de Circuitos 15-06-2021

P1)

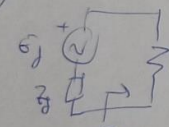
a)



$$P_e = \frac{1}{2}\left|\frac{\epsilon_g}{Z_g + Z_e}\right|^2 R_e[Z_e] = \frac{1}{2}\left|\frac{4}{2+3j+2-3j}\right|^2 \cdot 2 = 1\,W$$

$$\Rightarrow \boxed{P_s = P_e \cdot 10^{-\frac{\alpha_T}{10}} = \frac{1}{4}P_e = \frac{1}{4}\,W} \qquad \left\| \alpha_T = 10\log_{10}\frac{P_e}{P_s} \Rightarrow \frac{P_e}{P_s} = 10^{\frac{\alpha_T}{10}} \right.$$

b)



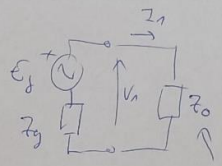
$$P_{2\omega} = \frac{1}{2}\left|\frac{\epsilon_g}{Z_g + R_L}\right|^2 \cdot R_L = \ldots = \frac{4}{9}\,W$$

$$\boxed{\alpha_I = 10\log\frac{P_{2\omega}}{P_2} = 10\log\frac{4^2}{9} = 2{,}5\,dB}$$

$$\left(\alpha_I = \frac{1}{2}\ln\frac{P_{2\omega}}{P_2} = 0{,}287\,Np\right)$$

P2)

a)



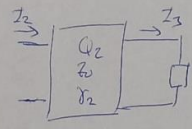
$$Z_o = \frac{V_1}{I_1}\,?$$

cal que $Z_L = Z_o$

$$\epsilon_g = Z_g I_1 = V_1 \Rightarrow I_1 = \frac{\epsilon_g - V_1}{Z_g} = (4+3j)\,A.$$

$$\Rightarrow \boxed{Z_o = \frac{10(7-j)}{4+3j} = 10(1-j)\,\Omega}$$

$$\boxed{\gamma_1 = \ln\left.\frac{I_1}{I_2}\right|_{Z_L = Z_o} = \ln\frac{4+3j}{\frac{4+3j}{10}} = \ln 10\,Np}$$



$$I_3 = \frac{V_2}{Z_o} = \frac{-7+j}{10(1-j)} = -0{,}4 - j0{,}3\,A$$

$$\boxed{\gamma_2 = \ln\frac{I_2}{I_3} = \ln\frac{4+3j}{-10(0{,}4+0{,}3j)} = \ln(-1) = \pm j\pi}$$

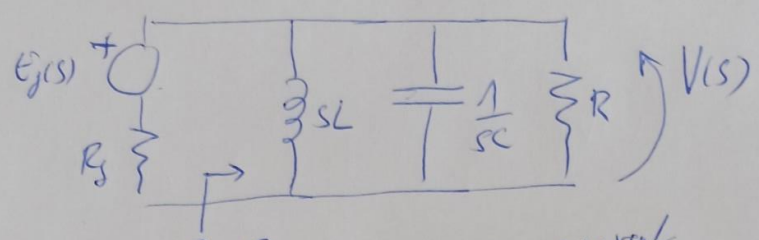
b)

$$\alpha_I(Q_2) = 0$$

$$\alpha_T(Q_1) = \ln(10)\,Np$$

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P3)



$R = R_g$ para $\omega = \omega_0 = 2\pi$ rad/s
se produce máxima transferencia de pot.

a)

$$\frac{V(s) - E_j(s)}{R_g} + \frac{V(s)}{s \cdot L} + V(s) \cdot s \cdot C + \frac{V(s)}{R} = 0$$

$$V(s)\left[\frac{1}{R_g} + \frac{1}{R} + \frac{1}{Ls} + Cs\right] = \frac{E_j(s)}{R_g}$$

$$\Rightarrow \boxed{H(s) = \frac{V(s)}{E_j(s)} = \frac{\frac{1}{R_g C} \cdot s}{s^2 + \frac{R+R_g}{RR_g \cdot C} s + \frac{1}{LC}}} = \frac{\frac{R}{R+R_g} \cdot \frac{R+R_g}{RR_g C} \cdot s}{s^2 + \frac{R+R_g}{RR_g C} s + \frac{1}{LC}}$$

K ; B_{3dB} ; ω_0^2

Filtro resonante pasabanda.

b) $\boxed{B_{3dB} = \frac{R+R_g}{RR_g \cdot C} \overset{R \doteq R_g}{=} 100 \text{ k rad/s}}$

$$K = \frac{R}{R+R_g} = \frac{1}{2} \overset{R = R_g}{=}$$

$$\omega_0^2 = \frac{1}{LC} \Rightarrow \boxed{C = \frac{1}{\omega_0^2 L} = 2\text{nF}}$$

$$\Rightarrow \boxed{Q = \frac{\omega_0}{B_{3dB}} = 20}$$

c) $V(s) = H(s) \cdot E_j(s)$, $e_j(t) = 4 + \cos(2 \cdot 10^6 t)$ V

($\omega = 0$; $\omega \doteq \omega_0$)

Para componente continua $\omega = 0 \Rightarrow H(s=0) = 0 \Rightarrow V(0) = 0$

" " $\omega = \omega_0 \Rightarrow H(\omega = \omega_0) = K = \frac{1}{2} \Rightarrow V(\omega_0) = \frac{1}{2} \cdot 1 = \frac{1}{2}$

$$\Rightarrow \boxed{v(t) = 0 + \frac{1}{2}\cos(2 \cdot 10^6 t) \text{ V}}$$

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